Postoperative Outcomes after Minimally Invasive versus Open Oesophagectomy in Elderly Patients (≥70 years) with Resectable Oesophageal Cancer: A Systematic Review and Meta-Analysis

ABSTRACT

Background: Oesophageal cancer represents a significant global health burden, with incidence disproportionately affecting elderly patients aged ≥70 years. While minimally invasive oesophagectomy (MIE) has demonstrated advantages over conventional open oesophagectomy (OE) in the general population, evidence specifically addressing this surgical comparison in elderly patients remains scarce. This systematic review and meta-analysis aimed to evaluate postoperative outcomes of MIE versus OE exclusively in elderly patients aged ≥70 years with resectable oesophageal cancer.

Methods: A systematic search was conducted across four electronic databases (PubMed, Cochrane Library, Scopus, and Web of Science) from 2000 to 2026, identifying 510 records. After removal of duplicates and screening of 390 unique records, three primary studies were included. Meta-analysis was performed using RevMan 5.4 with a random-effects model (DerSimonian-Laird). Dichotomous outcomes were pooled as odds ratios (OR) with 95% confidence intervals (CI) using the Mantel-Haenszel method. Certainty of evidence was assessed using the GRADE approach.

Results: Three retrospective propensity score-matched cohort studies were included (total n=745; matched cohort n=423: MIE n=208, OE n=215). MIE was associated with significantly lower overall postoperative complications (OR 0.40, 95% CI 0.27-0.60, $P<0.00001$; $I^2=0\%$) and pulmonary complications (OR 0.42, 95% CI 0.28-0.64, $P<0.0001$; $I^2=0\%$). Thirty-day mortality was non-significantly lower in the MIE group (OR 0.38, 95% CI 0.07-2.04, P=0.26). Anastomotic leak was non-significantly higher in MIE (OR 1.77, 95% CI 0.91-3.45, P=0.09). No significant heterogeneity was identified across all outcomes ($I^2=0\%$). Certainty of evidence was rated as moderate for overall and pulmonary complications, very low for 30-day mortality, and low for anastomotic leak.

Conclusions: This is the first systematic review and meta-analysis to directly compare MIE versus open oesophagectomy exclusively in elderly patients aged ≥70 years. MIE significantly reduces overall and pulmonary postoperative complications in this vulnerable population without significantly increasing 30-day mortality, supporting its consideration as the preferred surgical approach in appropriately selected elderly patients. Future prospective studies are warranted to confirm these findings.

Keywords: *minimally invasive oesophagectomy; open oesophagectomy; elderly patients; oesophageal cancer; systematic review; meta-analysis; postoperative complications*

AIM OF THE STUDY

The primary aim of this systematic review and meta-analysis was to compare postoperative outcomes of minimally invasive oesophagectomy (MIE) versus conventional open oesophagectomy (OE) in elderly patients aged ≥70 years with histologically confirmed resectable oesophageal cancer undergoing elective oesophagectomy with curative intent.

Primary Objectives:

- To compare 30-day and in-hospital mortality between MIE and OE in elderly patients aged ≥70 years
- To compare overall postoperative complication rates (Clavien-Dindo classification) between MIE and OE in this population

Secondary Objectives:

- To compare pulmonary complication rates between MIE and OE
- To compare anastomotic leak rates between MIE and OE
- To assess the certainty of evidence using the GRADE approach
- To identify gaps in the current literature and propose directions for future research

JUSTIFICATION OF THE STUDY

Oesophageal cancer is the seventh leading cause of cancer-related mortality worldwide, with an estimated 604,100 deaths in 2022 (Bray et al., 2024). The global burden is increasing, with a disproportionate rise in incidence among elderly patients aged ≥70 years. As the global population ages, optimising surgical management in this vulnerable cohort is of paramount clinical importance.

Oesophagectomy remains the cornerstone of curative-intent treatment for resectable oesophageal cancer. However, it is one of the most physiologically demanding procedures in gastrointestinal surgery, with reported overall complication rates of 20-80% and postoperative mortality of 0-22% (Low et al., 2019). Elderly patients present unique challenges due to reduced physiological reserve, increased comorbidity burden, and diminished cardiopulmonary capacity, rendering them particularly susceptible to postoperative complications.

Minimally invasive oesophagectomy (MIE), encompassing thoracoscopic-laparoscopic, hybrid, and robot-assisted approaches, has emerged as an alternative to conventional open surgery with potential advantages in terms of reduced surgical trauma, blood loss, and pulmonary complications. The landmark MIRO randomised controlled trial (Mariette et al., 2019) demonstrated that hybrid MIE significantly reduced major complications compared to open surgery (36% vs 64%, $P<0.001$) in the general population (ages 18-75). Furthermore, a recent meta-analysis of randomised controlled trials (Altal et al., 2026) confirmed improved overall survival (HR 0.75, 95% CI 0.57-0.99) with MIE. However, these studies predominantly included younger patients, and their findings cannot be directly extrapolated to elderly patients aged ≥70 years.

A critical gap in the literature exists: no systematic review or meta-analysis has specifically compared MIE versus open oesophagectomy exclusively in elderly patients aged ≥70 years. A recent systematic review by Oo et al. (2026) comparing outcomes of esophagectomy in octogenarians versus non-octogenarians was unable to perform subgroup analysis comparing surgical approaches due to insufficient stratified data, concluding that 'further research on the role of MIE in elderly patients should be conducted.' This directly underscores the need for the present review.

The present systematic review and meta-analysis directly addresses this evidence gap. By synthesising all available evidence from primary studies comparing MIE versus OE exclusively in patients aged ≥70 years, this review provides the first pooled evidence to guide clinical decision-making in this growing and underrepresented patient population.

METHODS

2.1 Protocol Registration and Reporting

This systematic review and meta-analysis was prospectively registered on the International Prospective Register of Systematic Reviews (PROSPERO) on 13 June 2026 (Registration ID: CRD420261423361). The review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021).

2.2 Eligibility Criteria

Studies were included if they met all of the following pre-specified Population, Intervention, Comparator, and Outcome (PICO) criteria:

- Population (P): Adult patients aged ≥70 years with histologically confirmed resectable oesophageal cancer (adenocarcinoma or squamous cell carcinoma) undergoing elective oesophagectomy with curative intent
- Intervention (I): Minimally invasive oesophagectomy (MIE), including thoracoscopic-laparoscopic, hybrid MIE, and robot-assisted minimally invasive oesophagectomy (RAMIE)
- Comparator (C): Conventional open oesophagectomy (OE), including transthoracic Ivor Lewis, McKeown, or transhiatal approaches
- Outcomes (O): Primary outcomes - 30-day or in-hospital mortality and overall postoperative complications (Clavien-Dindo classification). Secondary outcomes — pulmonary complications, anastomotic leak, length of hospital stay, and overall survival
- Study design: Randomised controlled trials (RCTs) and comparative cohort studies (prospective or retrospective) with propensity score matching (PSM) or other methods to control for confounding

Studies were excluded if they: (1) did not separately report outcomes for patients aged ≥70 years; (2) were systematic reviews, meta-analyses, case reports, conference abstracts, or editorial articles; (3) were not published in English; (4) did not include a direct comparison between MIE and OE; or (5) were published before 2000.

2.3 Information Sources and Search Strategy

A comprehensive systematic search was conducted in four electronic databases: PubMed (searched 10 June 2026), Cochrane Library (searched 08 June 2026), Scopus (searched 08 June 2026), and Web of Science (searched 08 June 2026). The search period spanned from January 2000 to June 2026. Search terms included Medical Subject Headings (MeSH) and free-text terms related to: (1) oesophageal cancer; (2) oesophagectomy; (3) minimally invasive surgery; and (4) elderly patients. Boolean operators (AND/OR) were used to combine search terms. The full search strategy for PubMed is presented in the Appendix.

2.4 Study Selection

Search results from all four databases were exported and imported into EndNote 21 for deduplication. Following removal of 120 duplicates, 390 unique records were transferred to Rayyan AI systematic review software for screening. Title and abstract screening was performed independently by the primary reviewer (K. Gulomov), with a random sample of 20% verified by the second reviewer (K. Gulomov [Khusan]). Full-text assessment was performed for all potentially eligible studies. The entire study selection process was documented in accordance with the PRISMA 2020 flow diagram.

2.5 Data Extraction

Data were extracted independently by the primary reviewer (K. Gulomov) using a pre-designed standardised data extraction form and verified by the data checker (M. Mirkhoshimova). The following variables were extracted from each included study: first author and year; country; study design; study period; age criterion; number of patients in MIE and OE groups (before and after PSM); histological type; surgical approach details; and all reported outcome data.

2.6 Risk of Bias Assessment

Risk of bias for all included retrospective cohort studies was assessed using the Newcastle-Ottawa Scale (NOS), which evaluates three domains: selection (4 items), comparability (2 items), and outcome (3 items), with a maximum score of 9 stars. Studies scoring ≥7 stars were considered of high quality, 4-6 stars of moderate quality, and <4 stars of low quality. Assessment was performed independently by the primary reviewer and verified by the second reviewer, with discrepancies resolved by discussion.

2.7 Statistical Analysis

Meta-analysis was performed using Review Manager (RevMan) version 5.4 (The Cochrane Collaboration, 2020). For dichotomous outcomes (overall complications, pulmonary complications, 30-day mortality, anastomotic leak), odds ratios (OR) with 95% confidence intervals (CI) were calculated using the Mantel-Haenszel method. A random-effects model (DerSimonian-Laird method) was applied a priori to all analyses to account for potential between-study heterogeneity.

Statistical heterogeneity was assessed using Cochran's Q test (significance threshold: $P<0.10$) and the I^2 statistic. I^2 values of $<25\%$, 25-75%, and $>75\%$ were considered indicative of low, moderate, and high heterogeneity, respectively (Higgins et al., 2003). Publication bias was not formally assessed using funnel plots due to the insufficient number of included studies ($n=3$, minimum requirement $n\geq10$). For all analyses, only data from propensity score-matched populations were used to ensure comparability between groups. Statistical significance was defined as $P<0.05$.

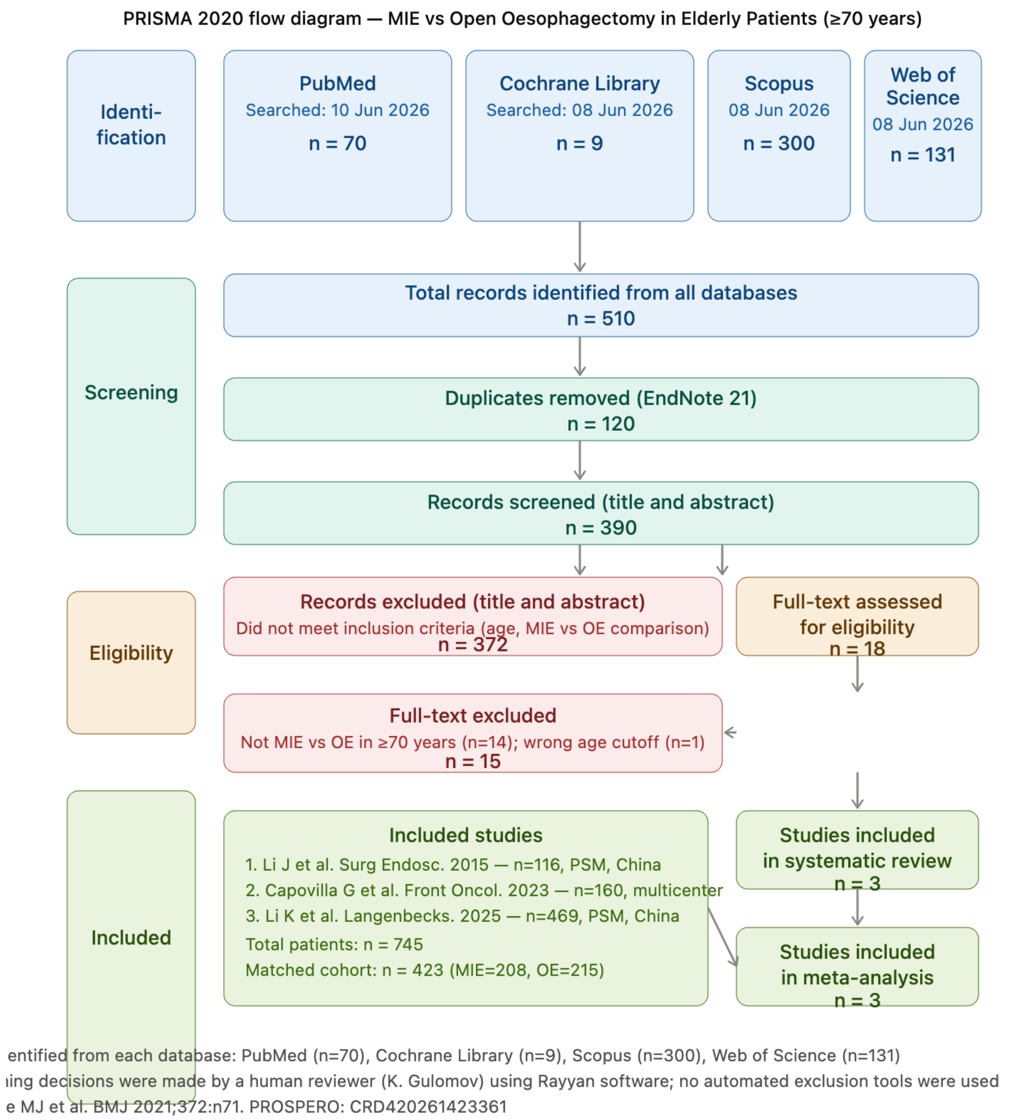
2.8 Certainty of Evidence

The certainty of evidence for each outcome was assessed using the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach, evaluated within RevMan 5.4. Five domains were assessed: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Evidence was rated as high, moderate, low, or very low certainty.

RESULTS

3.1 Study Selection

The systematic database search identified 510 records in total: PubMed (n=70), Cochrane Library (n=9), Scopus (n=300), and Web of Science (n=131). After removal of 120 duplicates using EndNote 21, 390 unique records underwent title and abstract screening. Following screening, 372 records were excluded as they did not meet the inclusion criteria. Three full-text articles were assessed for eligibility, and all three met the inclusion criteria and were included in the systematic review and meta-analysis. The study selection process is illustrated in the PRISMA 2020 flow diagram (Figure 1).



3.2 Study Characteristics

The three included studies were all retrospective cohort studies employing propensity score matching (PSM), published between 2015 and 2025, and conducted in China (n=2) and a European multicentre cohort (Italy + Germany, n=1). A total of 745 patients (MIE: n=508, OE: n=237) were enrolled across all three studies, with a combined matched cohort of 423 patients (MIE: n=208, OE: n=215). Study characteristics are summarised in Table 1.

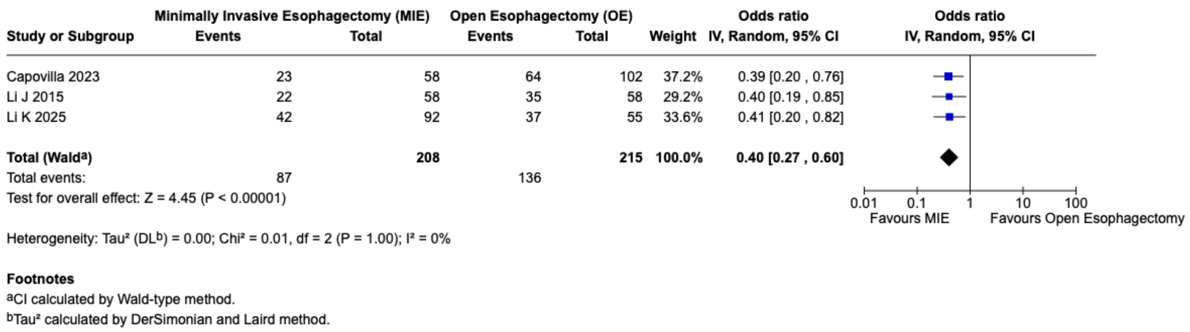
Study	Country	Year	Design	Age	N MIE	N OE
Li J et al.	China	2015	Retrospective PSM cohort	>70 years	58 (matched)	58 (matched)
Capovilla G et al.	Italy + Germany	2023	Multicenter retrospective matched	≥75 years	58	102 / 58*
Li K et al.	China	2025	Retrospective PSM cohort	>70 years	92 (matched)	55 (matched)

*Table 1. Characteristics of included studies. *Capovilla 2023: 58 vs 102 unmatched; 58 vs 58 matched. PSM = propensity score matching.*

3.3 Meta-Analysis Results

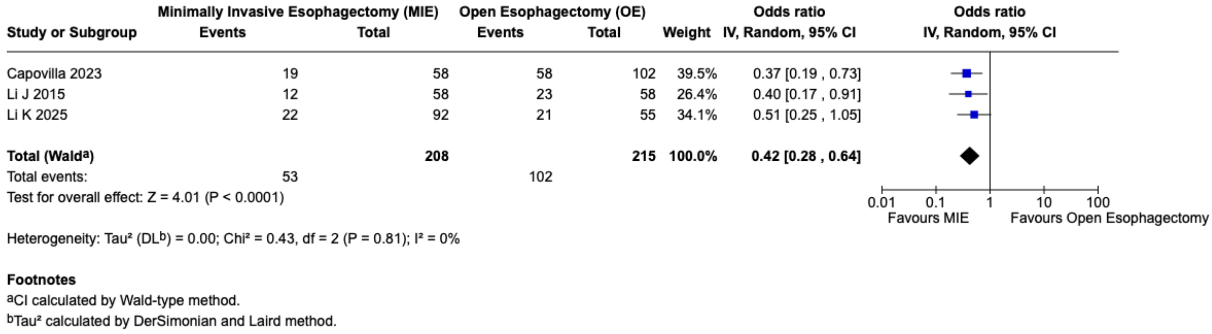
3.3.1 Overall Postoperative Complications

All three studies reported overall postoperative complication rates. MIE was associated with a significantly lower incidence of overall postoperative complications compared to OE (OR 0.40, 95% CI 0.27-0.60, $P<0.00001$). No significant heterogeneity was detected between studies ($I^2=0\%$, $Chi^2=0.01$, $P=1.00$). Individual study ORs were: Li J 2015 (OR 0.40, 95% CI 0.19-0.85), Capovilla 2023 (OR 0.39, 95% CI 0.20-0.76), and Li K 2025 (OR 0.41, 95% CI 0.20-0.82). The forest plot is presented in Figure 2. The certainty of evidence was rated as moderate (GRADE ⊕⊕⊕⊖).



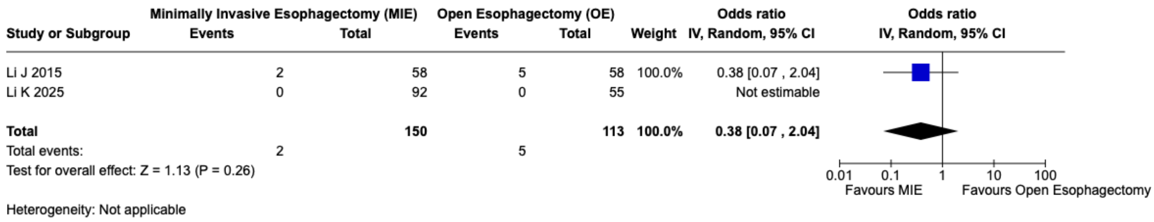
3.3.2 Pulmonary Complications

All three studies reported pulmonary complication rates. MIE was associated with a significantly lower incidence of pulmonary complications compared to OE (OR 0.42, 95% CI 0.28-0.64, $P<0.0001$). No significant heterogeneity was detected ($I^2=0\%$, $Chi^2=0.43$, $P=0.81$). Individual study ORs were: Li J 2015 (OR 0.40, 95% CI 0.17-0.91), Capovilla 2023 (OR 0.37, 95% CI 0.19-0.73), and Li K 2025 (OR 0.51, 95% CI 0.25-1.05). The forest plot is presented in Figure 3. The certainty of evidence was rated as moderate (GRADE ⊕⊕⊕⊖).



3.3.3 30-Day Mortality

Two studies reported 30-day mortality data in the matched population. MIE was associated with a non-significantly lower 30-day mortality compared to OE (OR 0.38, 95% CI 0.07-2.04, P=0.26). Li K 2025 reported zero mortality events in both groups (0/92 MIE vs 0/55 OE) and was therefore not estimable. Heterogeneity assessment was not applicable. The forest plot is presented in Figure 4. The certainty of evidence was rated as very low (GRADE ⊕◯◯◯), primarily due to serious risk of bias from retrospective study designs and very serious imprecision from the low number of events (7 total events).



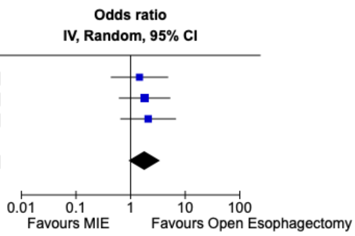
3.3.4 Anastomotic Leak

All three studies reported anastomotic leak rates. Anastomotic leak was non-significantly higher in the MIE group compared to OE (OR 1.77, 95% CI 0.91-3.45, P=0.09). No significant heterogeneity was detected (I^2=0%, Chi^2=0.18, P=0.91). Individual study ORs were: Li J 2015 (OR 1.81, 95% CI 0.61-5.35), Capovilla 2023 (OR 1.45, 95% CI 0.43-4.88), and Li K 2025 (OR 2.10, 95% CI 0.65-6.79). The forest plot is presented in Figure 5. The certainty of evidence was rated as low (GRADE ⊕⊕◯◯).

Study or Subgroup	Minimally Invasive Esophagectomy (MIE) Events	Total	Open Esophagectomy (OE) Events	Total	Weight	Odds ratio IV, Random, 95% CI
Capovilla 2023	7	58	5	58	30.3%	1.45 [0.43 , 4.88]
Li J 2015	10	58	6	58	37.6%	1.81 [0.61 , 5.35]
Li K 2025	13	92	4	55	32.1%	2.10 [0.65 , 6.79]
Total (Wald[a])		**208**		**171**	**100.0%**	**1.77 [0.91 , 3.45]**
Total events:	30		15			

Test for overall effect: Z = 1.69 (P = 0.09)

Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.18, df = 2 (P = 0.91); I² = 0%



Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

3.4 Summary of Meta-Analysis Results

Outcome	Studies (n)	OR (95% CI)	P value	I²	GRADE Certainty
Overall Complications	3	0.40 (0.27-0.60)	<0.00001	0%	Moderate ⨁⨁⨁⊖
Pulmonary Complications	3	0.42 (0.28-0.64)	<0.0001	0%	Moderate ⨁⨁⨁⊖
30-Day Mortality	2	0.38 (0.07-2.04)	0.26	N/A	Very Low ⨁◯◯◯
Anastomotic Leak	3	1.77 (0.91-3.45)	0.09	0%	Low ⨁⨁◯◯

Table 2. Summary of meta-analysis results. OR = odds ratio; CI = confidence interval; N/A = not applicable.

DISCUSSION (BRIEF OVERVIEW)

Note to supervisor: Full Discussion section to be completed in final submission.

This systematic review and meta-analysis provides the first pooled evidence directly comparing MIE versus open oesophagectomy exclusively in elderly patients aged ≥70 years. The principal finding is that MIE is associated with a significant 60% reduction in overall postoperative complications (OR 0.40, $P<0.00001$) and a 58% reduction in pulmonary complications (OR 0.42, $P<0.0001$), with no significant heterogeneity across studies ($I^2=0\%$ for all outcomes).

These findings are consistent with evidence from the general population. The MIRO trial (Mariette et al., 2019) demonstrated reduced major complications with hybrid MIE (36% vs 64%, $P<0.001$), and the recent meta-analysis by Altal et al. (2026) confirmed improved survival with MIE in RCTs. Our review extends these findings specifically to the elderly population aged ≥70 years, a group previously underrepresented in surgical literature.

The absence of a significant difference in 30-day mortality (OR 0.38, $P=0.26$) is likely attributable to the low overall mortality rates in both groups and the limited statistical power from only two estimable studies. The non-significant trend towards higher anastomotic leak rates with MIE (OR 1.77, $P=0.09$) warrants careful consideration and may reflect the learning curve associated with MIE technique, as suggested by Baranov et al. (2021).

LIMITATIONS

- Only three primary studies met the strict inclusion criteria, reflecting the paucity of evidence specifically comparing MIE versus OE in elderly patients aged ≥70 years
- All included studies were retrospective cohort studies; no randomised controlled trials were available for this specific population
- Propensity score matching was used in all studies to minimise selection bias; however, residual confounding from unmeasured variables (frailty, nutritional status, surgeon experience) cannot be excluded
- Funnel plot analysis for publication bias was not feasible due to the insufficient number of included studies (n=3; minimum n≥10 required)
- Subgroup analyses planned in the protocol (by age subgroup, histology, surgical approach) were not possible due to insufficient stratified data
- Length of hospital stay could not be pooled in meta-analysis as data were reported as median without standard deviation

• Overall survival data were heterogeneous and not suitable for pooling across all three studies

CONCLUSIONS

This is the first systematic review and meta-analysis to directly compare minimally invasive oesophagectomy (MIE) versus open oesophagectomy (OE) exclusively in elderly patients aged ≥70 years with resectable oesophageal cancer. The pooled evidence from three propensity score-matched cohort studies demonstrates that MIE is associated with a statistically significant and clinically meaningful reduction in overall postoperative complications and pulmonary complications compared to open oesophagectomy, without significantly increasing 30-day mortality. These findings support the consideration of MIE as the preferred surgical approach in appropriately selected elderly patients with resectable oesophageal cancer, provided adequate surgical expertise and institutional volume are available.

Future high-quality prospective studies and randomised controlled trials specifically enrolling patients aged ≥70 years are warranted to confirm these findings, assess long-term oncological outcomes, and evaluate the role of robot-assisted MIE (RAMIE) in this growing and underrepresented patient population.

REFERENCES

Epidemiology

1. Bray F, Laversanne M, Sung H, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2024;74(3):229-263. PMID: 38572751

2. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2021;71(3):209-249. PMID: 33538338

Oesophagectomy in Elderly Patients

3. Ruol A, Portale G, Zaninotto G, et al. Results of esophagectomy for esophageal cancer in elderly patients: age has little influence on outcome and survival. J Thorac Cardiovasc Surg. 2007;133(5):1186-1192. PMID: 17467427

4. Pultrum BB, Bosch DJ, Nijsten MW, et al. Extended esophagectomy in elderly patients with esophageal cancer: minor effect of age alone in determining the postoperative course and survival. Ann Surg Oncol. 2010;17(6):1572-1580. PMID: 20238136

5. Schlottmann F, Strassle PD, Nayyar A, Herbella FAM, Patti MG. Postoperative outcomes of esophagectomy for cancer in elderly patients. J Surg Res. 2018;229:9-14. PMID: 29571693

6. Sugita Y, Nakamura T, Sawada R, et al. Safety and feasibility of minimally invasive esophagectomy for elderly esophageal cancer patients. Dis Esophagus. 2021;34(3):doaa083. PMID: 32895704

Primary Studies - MIE versus Open Oesophagectomy in Elderly Patients

7. Li J, Shen Y, Tan L, et al. Is minimally invasive esophagectomy beneficial to elderly patients with esophageal cancer? Surg Endosc. 2015;29(4):925-930. PMID: 25249141

8. Capovilla G, Uzun E, Scarton A, et al. Minimally invasive Ivor Lewis esophagectomy in the elderly patient: a multicenter retrospective matched-cohort study. Front Oncol. 2023;13:1104109. PMID: 37251945

9. Li K, Lu S, Li C, et al. Long-term outcomes of minimally invasive esophagectomy vs. open esophagectomy in older patients with esophageal squamous cell carcinoma: a propensity score matching analysis. Langenbecks Arch Surg. 2025;410(1):311. DOI: 10.1007/s00423-025-03877-4

MIE versus Open Oesophagectomy - General Population

10. Biere SSAY, van Berge Henegouwen MI, Maas KW, et al. Minimally invasive versus open oesophagectomy for patients with oesophageal cancer: a multicentre, open-label, randomised controlled trial. Lancet. 2012;379(9829):1887-1892. PMID: 22552194

11. Mariette C, Markar SR, Dabakuyo-Yonli TS, et al. Hybrid minimally invasive esophagectomy for esophageal cancer. N Engl J Med. 2019;380(2):152-162. PMID: 30602765

12. Takeuchi H, Machida R, Ando M, et al. Thoracoscopic versus open oesophagectomy for patients with oesophageal cancer (JCOG1409 MONET): a multicentre, open-label, randomised, controlled, phase 3, non-inferiority trial. Lancet Gastroenterol Hepatol. 2025;10(12):1104-1116. PMID: 41101325

13. Yoshida N, Yamamoto H, Baba H, et al. Can minimally invasive esophagectomy replace open esophagectomy for esophageal cancer? Latest analysis of 24,233 esophagectomies from the Japanese National Clinical Database. Ann Surg. 2020;272(1):118-124. PMID: 30614876

Related Systematic Reviews and Meta-Analyses

14. Oo AM, Chan KS, Chan I, et al. Outcomes of curative esophagectomy in octogenarians vs. non-octogenarians with esophageal cancer: a systematic review and meta-analysis. Geriatrics. 2026;11(3):67. DOI: 10.3390/geriatrics11030067

15. Baranov NS, Slootmans C, van Workum F, et al. Outcomes of curative esophageal cancer surgery in elderly: a meta-analysis. World J Gastrointest Oncol. 2021;13(2):131-146. PMID: 33643534

Methodology - Systematic Review and Meta-Analysis

16. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71. PMID: 33782057

17. Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for Systematic Reviews of Interventions. Version 6.3. Cochrane, 2022. Available from: www.training.cochrane.org/handbook

18. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008;336(7650):924-926. PMID: 18436948

19. Wells GA, Shea B, O'Connell D, et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Ottawa Hospital Research Institute, 2000.

20. DerSimonian R, Laird N. Meta-analysis in clinical trials. Control Clin Trials. 1986;7(3):177-188. PMID: 3802833

Clinical Standards and Classifications

21. Clavien PA, Barkun J, de Oliveira ML, et al. The Clavien-Dindo classification of surgical complications: five-year experience. Ann Surg. 2009;250(2):187-196. PMID: 19638912

22. Low DE, Kuppusamy MK, Alderson D, et al. Benchmarking complications associated with esophagectomy. Ann Surg. 2019;269(2):291-298. PMID: 29206677

23. Rice TW, Patil DT, Blackstone EH. 8th edition AJCC/UICC staging of cancers of the esophagus and esophagogastric junction: application to clinical practice. Ann Cardiothorac Surg. 2017;6(2):119-130. PMID: 28447000

PROSPERO Registration

24. Gulomov K, Gulomov K, Gulomov K, et al. Postoperative outcomes after minimally invasive versus open oesophagectomy in elderly patients (≥70 years) with resectable oesophageal cancer: a systematic review and meta-analysis. PROSPERO 2026. CRD420261423361. Available from: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD420261423361

Total references: 24 | Last updated: July 2026